

Quick Guide – Using LaTeXML to convert LaTeX files

Using LaTeXML to convert LaTeX files to HTML

This guide provides instructions on installing and using LaTeXML to convert LaTeX files into HTML.

Installation

LaTeXML is available in many package managers for Linux, Mac OS, or through Perl's CPAN. You can find out more on [the official installation page](#), or find more detailed instructions for Windows [on StackExchange](#).

Running LaTeXML

The Basics

Turning a TeX file into a HTML file with LaTeXML is a two-stage process: TeX files are processed into an internal XML representation, and then into HTML for output¹. You should not need to edit the xml directly, although this may be necessary in extreme cases.

In the following examples, `notes.tex` is the tex file that needs converted. Any additional, e.g. images, are in the same folder.

Ensure that current directory of the terminal you are working in has been switched to this folder.

First, run the command `latexml` to compile the xml file:

```
latexml notes.tex --dest=notes.xml
```

This command produces a lot of textual output and may take some time if your document is large or uses a lot of macros or packages (our 100-pages example took 20 minutes to compile).

This will create a new file in the same folder with the name `notes.xml`.

Next, we produce the HTML file with the command:

```
latexmlpost notes.xml --dest=notes-folder/notes.html --split  
--splitat=section --navigationtoc=context --javascript="
```

¹ It is possible to convert directly from LaTeX to HTML using the command `latexmlc`, but we will not do this here. One reason it is useful to have the XML around is to allow you to play around with the options to `latexmlpost` without having to rerun the lengthy `latexml` command.

Quick Guide – Using LaTeXML to convert LaTeX files

```
https://cdnjs.cloudflare.com/ajax/libs/mathjax/2.7.9/MathJax.js?config=TeX-MML-AM_CHTML" --urlstyle=file --timestamp=0
```

This is a single command that should be typed in one line. It will run much faster than the previous one. The full options are detailed in the next section. This command instructs `latexmlpost` to create a new folder called `notes-folder` which will contain the file `notes.html` and any files it depends on. We are also instructing it to include [MathJax](#) for rendering mathematics in a screen reader friendly way. The entire contents of `notes-folder` should be accessible when you make this available on the internet².

The Details

While the above instructions should be enough for basic usage, you may find you need to customise certain behaviours. There is no substitute for [reading the documentation](#), but we will summarise some of the most common ones below.

Latexml

- `--dest=notes.xml`

Like most Unix programs, `latexml` defaults to printing out the document on “standard output” (that is, to the terminal) rather than to a file. With this option, it outputs the xml to the file `notes.xml`

- `--includestyles`

This allows `latexml` to load raw `.sty` files, which can be useful for loading custom class files.

Latexmlpost

- `--dest=notes.html`

As above, this tells `latexmlpost` to output the html to the file `notes.html`. Any additional files necessary for rendering the HTML will be put in the same folder, so it is often sensible to specify a folder as well as filename, e.g., `--dest=notes-folder/notes.html`

² In the simplest case, deployment of these files is just copy/pasting to an accessible location on a web server, but for more detailed deployment you will need to talk to your system administrator.

Quick Guide – Using LaTeXML to convert LaTeX files

- `--split`

This tells `latexmlpost` to split the output into several HTML pages, as opposed to one document. For shorter documents this is unnecessary, but desirable for large documents to cut down html loading times.

- `--splitat=subsection`

Used in combination with the above option, this tells `latexmlpost` which level of LaTeX heading it splits into a new page, e.g. section or subsection.

- `--navigationtoc=context`

This generates a table of contents in a navigation bar on each page.

- `--javascript=js-file`

This includes a link to the javascript file `js-file`, which may be a local file or a link to another web page³. We are primarily using it to include MathJax for rendering mathematics.

- `--css=css-file`

As above, but for CSS.

- `--urlstyle=file`

This determines the style of link used by LaTeXML.

- `--timestamp=tstamp`

Includes a specific timestamp for when the file is generated in the comments of the HTML file. By default `latexmlpost` uses current date and time, or you can use 0 to omit the timestamp.

General Advice

Before going on to explain how to fix common issues that occur during the conversion process, here is a short list of things that might make your life easier.

1. For large documents, we recommend to use the `/input` and `/include` LaTeX commands to split the content on several tex files. It makes it easier to selectively include /

³ This is not the time and place for a Javascript/CSS Tutorial, but to get an example of what can be done, you should look at [andy-navbar.js](#) and [andy-navbar.css](#) in this tutorial.

Quick Guide – Using LaTeXML to convert LaTeX files

exclude pieces of the notes while you debug any formatting issues and will have faster compilation times.

2. If you want things to be done a certain way depending on whether generating PDF or HTML, you can take advantage of the command `\iflatexml \else \fi`.
3. Ask yourself if you really need to bring in an additional library just for one small improvement. LaTeXML does not support every LaTeX package, and there is no (and can't be) a comprehensive list of those supported vs not. This includes some core packages, like [mdframed](#), which is used for coloured boxes, and [imakeidx](#) for creating indexes. Also [LaTeXML points out that some low-level TeX primitives are not mimicked faithfully](#) (see next section).
4. Since the purpose of this document is to create accessible documents, you should ensure that you are doing other necessary steps, e.g., checking if your images are colour-blind friendly⁴, and providing accurate and useful alt-text for your images using the `\description` command⁵.
5. Similarly, to take full advantage of HTML's ability to reflow text, you should make limited use of commands that specify precise spacing. Some commands like `\hfill` no longer make sense in a web context where there is no inherent notion of page width.

Formatting Issues

Assuming there were no significant errors in the `latexml` or `latexmlpost` phases, you should have got one or more HTML files. However, unless you were quite conservative in which LaTeX packages and features you used, there are likely to be some formatting issues to correct. It is impossible to be comprehensive here, but we list below some errors we found and how to fix those.

See also the following website from the [University of Leicester for LaTeXML bugs and fixes](#).

⁴ A summary of the issues and some suggestions can be found at [Tools to help you make colorblind friendly graphics](#).

⁵ The command `\Description[<short description>]{<long description>}` can, and should, be used inside every figure and table. It differs from `\caption` in that it is meant to be used instead of the image, rather than in combination with it.

Quick Guide – Using LaTeXML to convert LaTeX files

[NOFIX] Align environment not handling empty lines

`\align*` environments are often used to provide sequences of equalities alongside a “running commentary” explaining the proof. While this does work correctly, when a line has only the final comment (see below), it is not being aligned correctly.

```
\begin{align*}
&& \text{\small{\color{red} \emph{now} subtract}}
\\
&&\text{\small{\color{red} finally, don't expand the
bottom}}
\end{align*}
```

Warning: This is likely a bug in LaTeXML, and we were unable to find a workaround. However, in many cases it will be straightforward to append the final comment to the previous line or add it after the `align*` environment.

Colour leaking to other parts of document

One issue in the calculus was the command `\book` as defined as follows

```
\newcommand{\book}[1]{\marginpar{\tiny\color{brown} Stewart
#1}}
```

This caused everything in the document after the first instance of `\book` to be tiny and brown, indicating that these commands were not constrained to the `\marginpar` environment. This seems to be a bug, but is easily fixed by using a second pair of braces to delimit the effect, e.g.,

```
\newcommand{\book}[1]{\marginpar{{\tiny\color{brown} Stewart
#1}}}
```

It is likely that any other “state leakage” can be contained in the same way.

Note: We do not recommend the use of `\marginpar` in HTML. It produces somewhat ugly output, and the text is only available when you hover over the cross (†) that indicates it.

Quick Guide – Using LaTeXML to convert LaTeX files

Set Builder Notation

It is common to typeset sets using so-called “set builder notation”, e.g., the set of all reals satisfying a predicate P might be typeset as

$$\left\{x \in \mathbb{R} \mid P(x)\right\}.$$

However, the use of `\middle` was not handled correctly. Fortunately, in this case you can replace it with `\mid` instead giving

$$\left\{x \in \mathbb{R} \mid P(x)\right\}.$$

[NOFIX] () in blank sections

When a `\section` command is followed by no text before the first `\subsection`, it is being rendered as a pair of brackets. We currently know of no workaround. This does not seem to exist in the xml file, so is being produced by `latexmlpost` and is likely a bug.

Tikz

Our own example makes heavy use of [Tikz](#), a popular library for creating vector graphics within LaTeX. While many diagrams were produced correctly, due to Tikz’s size and complexity, it ended up being responsible for the lion’s share of the issues in the formatting of the notes.

Unfortunately, this was not limited to producing incorrect diagrams, but also sometimes caused bits of the document to be missing, or out of place. In general, if you are using Tikz and a formatting issue arises near a Tikz image, try commenting it out and see if the image was causing it.

While we will list the fixes we know for some of the issues we found, we recommend converting your image descriptions into image files, e.g. in SVG format, and including them as you would a regular image. A brief tutorial for this is provided at the end of this section.

Multiple Images side by side

A single math mode expression can contain multiple `tikzpicture` environments side by side, e.g.,

$$\begin{tikzpicture} \dots \end{tikzpicture} \dots \begin{tikzpicture} \dots \end{tikzpicture} \dots$$

This gets displayed as an error message containing “Math Input Error”. To fix this you can take advantage of the [subcaption](#) package to create a figure containing several subfigures and have a single tikz expression in each subfigure. An example can be found [on stackexchange](#).

Quick Guide – Using LaTeXML to convert LaTeX files

Converting TikZ Code to Images

If you are having significant issues with TikZ, you might consider converting all your TikZ images to standalone images. This can be accomplished by creating standalone pdfs with your image⁶ and then converting with a variety of programs including [ImageMagick](#), [pdf2cairo](#), [pdf2image](#), [pdf2svg](#), or some web converter (you'll find tons if you google).

For example⁷, if you have the following latex in `example.tex`

```
\documentclass[tikz, border=1mm]{standalone}

\begin{document}

\begin{tikzpicture}

\draw (0,0) node [] {My text};

\end{tikzpicture}

\end{document}
```

Compiling the latex with `pdflatex example.tex` will produce the file `example.pdf` which you can convert to, e.g., SVG, with `pdf2svg example.pdf example.svg`. Finally, you would include the image in your LaTeX code as normal with `\includegraphics{example.svg}`.

Further Resources

- [Blackboard Help](#)

⁶ This is strictly speaking unnecessary since you can include PDFs as images, but it is better practice since we want the browser to render it as an image rather than including a PDF.

⁷ This example was taken from [How to export tikz graphics as svg](#)